

UNIT – 1

Q1. Types of Operating System

Definition: OS manages hardware/software resources and provides services. **Types:** 1. Batch OS 2. Multiprogramming OS 3. Multitasking OS 4. Time-Sharing OS 5. Real-Time OS 6. Distributed OS 7. Network OS 8. Mobile OS **Real-Life Example:** Using Android phone while streaming & chatting. **Syllabus Example:** Time-Sharing OS, Real-Time OS. **Conclusion:** Types of OS improve efficiency for different tasks.

Q2. Components and Services of OS

Components: Process Mgmt, Memory Mgmt, File Mgmt, I/O Mgmt, Security, Shell. **Services:** Program Execution, I/O Ops, File Manipulation, Error Detection, Resource Allocation. **Real-Life Example:** Printing a document smoothly. **Syllabus Example:** OS architecture & kernel functions. **Conclusion:** Components and services ensure smooth system operation.

Q3. Buffering, Spooling, and Caching

Definition: Techniques to improve I/O efficiency. **Buffering:** Temporary memory for data transfer. **Spooling:** Queues data for sequential device access. **Caching:** Stores frequently used data for quick access. **Real-Life Example:** Streaming video, print queues, browser cache. **Syllabus Example:** I/O Management. **Conclusion:** Improves data transfer and system performance.

UNIT – 2

Q1. Process Management

Definition: Process = program in execution. **Functions:** Creation, Scheduling, Synchronization, IPC, Deadlock Handling. **Real-Life Example:** WhatsApp handling messages and notifications. **Conclusion:** Ensures efficient execution and resource allocation.

Q2. TCB / PCB

Definition: PCB stores all info about a process. **Contents:** PID, State, Program Counter, CPU Registers, Memory Info, Accounting, I/O Status. **Real-Life Example:** Pausing/resuming a game. **Conclusion:** PCB = process identity card.

Q3. Process State Transition Diagram

States: New, Ready, Running, Waiting, Terminated. **Real-Life Example:** Video loading → ready → running → buffering → terminated. **Conclusion:** Shows process life cycle & CPU management.

Q4. Schedulers

Definition: Decide which process runs next. **Types:** Long-Term, Short-Term, Medium-Term. **Real-Life**

Example: Suspending background apps on phone. **Conclusion:** CPU time is allocated efficiently.

Q5. FCFS with Example

Definition: Non-preemptive, first-arrival executed first. **Example:** P1(4), P2(3), P3(1) → Gantt: P1|P2|P3.

Real-Life Example: Queue at movie ticket counter. **Conclusion:** Simple but may delay short processes.

Q6. Threads

Definition: Smallest execution unit within a process. **Types:** User-level, Kernel-level. **Real-Life Example:**

Each Chrome tab as a thread. **Conclusion:** Improves parallelism and responsiveness.

Q7. Scheduling Criteria

Definition: Parameters to evaluate CPU scheduling. **Criteria:** CPU Utilization, Throughput, Turnaround, Waiting Time, Response Time, Fairness. **Conclusion:** Ensures efficient process scheduling.

UNIT – 3

Q1. Fragmentation and Types

Definition: Memory inefficiently used, leaving holes. **Types:** Internal (inside allocated block), External (small scattered free blocks). **Real-Life Example:** Books on a shelf not fitting exactly. **Conclusion:** Reduces memory efficiency; paging solves it.

Q2. Paging / Demand Paging

Definition: Divides memory into frames/pages; loads page only when needed. **Steps:** Check memory → page fault → load page → update page table. **Real-Life Example:** Opening large PDF, only load scrolled pages. **Conclusion:** Efficient memory usage, avoids fragmentation.

Q3. Page Replacement Algorithm (FIFO)

Definition: Removes oldest page when memory full. **Steps:** Maintain queue → on page fault remove front → insert new page. **Example:** Page references 1,2,3,4,1 in 3 frames. **Real-Life Example:** Queue in cafeteria. **Conclusion:** Simple but can cause Belady's anomaly.

UNIT – 4

Q1. 5 Basic Commands

Commands: pwd, ls, cd, mkdir, rm. **Real-Life Example:** Navigating folders on phone. **Conclusion:** CLI interaction made efficient.

Q2. Process Management Commands

Commands: ps, top/htop, kill, nice/renice, jobs/fg/bg. **Real-Life Example:** Closing frozen app with kill. **Conclusion:** Monitors and controls processes effectively.

Q3. Directory Structure / Access Methods

Structures: Single-level, Two-level, Tree, Acyclic Graph. **Access Methods:** Sequential, Direct, Indexed. **Real-Life Example:** Phone folder tree; direct photo search. **Conclusion:** Organizes files efficiently.

Q4. UNIX and Features

Definition: Multiuser, multitasking OS from 1969. **Features:** Multiuser, Multitasking, Portability, Hierarchical FS, Security, Shell, Pipes/Filters. **Real-Life Example:** Servers hosting websites. **Conclusion:** Robust, secure, flexible OS.

 All Units compiled for **quick revision and exam preparation.**